UNS

From PLC to ERP: how vNode builds your Unified Namespace

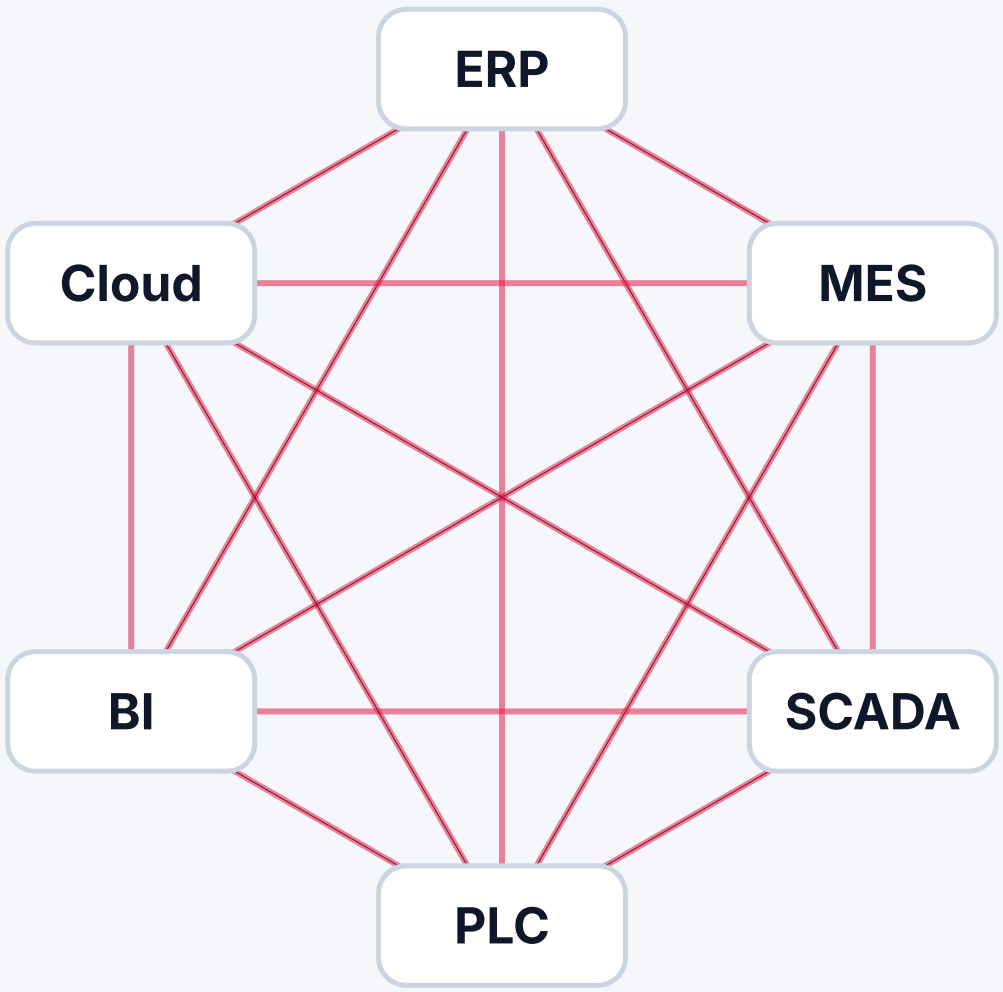
What a UNS is, why it matters and where vNode fits, in 8 slides.

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THE PROBLEM

Point-to-point integrations

Every system connects to every other one. With just 6 systems that is up to 15 custom integrations.



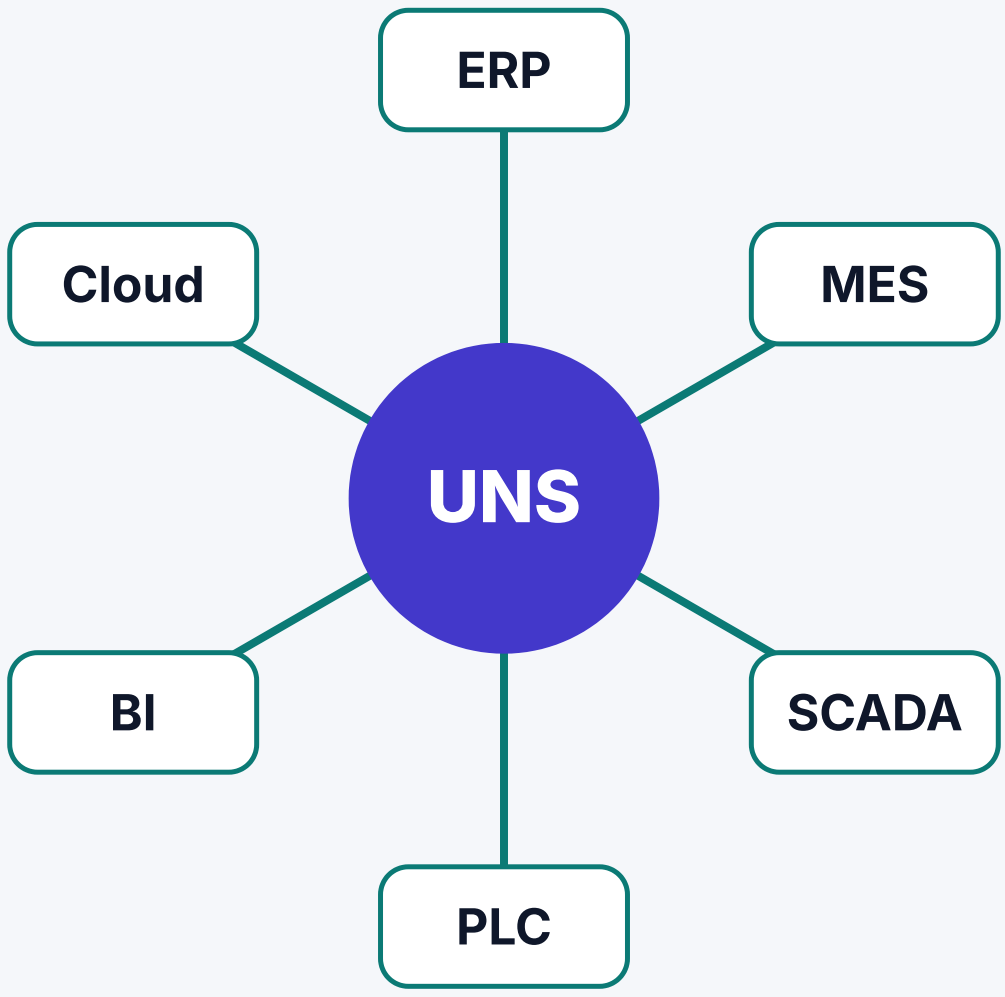
- × Expensive to build and maintain
- × Fragile: one change breaks others
- × Data stuck in silos

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THE SOLUTION

One central hub: the UNS

Each system connects only once: it publishes what it knows and subscribes to what it needs.



Publish/subscribe
Producers and consumers decoupled

ISA-95 hierarchy
Every data point in its semantic place

Current state
A single real-time source of truth

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1 vNode Edge

Connects to the plant floor and adds context

Native OT drivers

S7, EtherNet/IP, Modbus, OPC UA, DNP3, IEC 104, IEC 61850, BACnet

ISA-95 templates

Reusable asset models

Sparkplug B and JSON

Standard publishing over MQTT

Store & Forward

No data loss when the connection drops

JavaScript at the edge

Calculations and normalization close to the data

Bidirectional

Reads from and writes to equipment

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2 vNode Central

Consumes, processes and delivers

Wildcard subscriptions

Listens to whole branches of the UNS

Data quality

Interprets payloads and respects their state

Republishes KPIs

Gives value back to the UNS

Delivery to IT

Historian, SQL, MES, ERP / SAP, AWS, Azure

Relays over other protocols

REST, OPC UA, Modbus, DNP3, IEC 104, files

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THE FULL ARCHITECTURE

From plant floor to business

Business (IT)
Historian · SQL · MES · ERP / SAP · AWS · Azure

⇅ delivery to IT

vNode Central
consumes, processes and delivers

⇅ MQTT · wildcards

MQTT Broker = UNS
`enterprise/site/area/line/asset/tag`

⇅ MQTT · event-driven

vNode Edge
connects and adds context

⇅ read and write

Plant floor (OT)
PLCs · RTUs · IEDs · sensors

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BENEFITS

Why it matters

End customer

- Lower integration and maintenance costs
- No vendor lock-in
- Data ready for analytics and AI
- Scale from one line to many sites

Integrator

- Faster, repeatable projects
- Less risk and less support
- Digital transformation partner
- Recurring services

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A UNS is an architecture, not a product

Technology helps, but what makes the difference is a solid data model and clear governance.

> **Already running a UNS, or still living with point-to-point integrations?**
>
> Tell me in the comments 👇